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MATHEMATICS

Mathematics: Sets, Functions and Beyond — Part 3

Class 11

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Chapter 1

Sets

A set is a well-defined collection of distinct objects. Sets are described in roster or set-builder form and can be finite or infinite. Operations on sets include union, intersection, difference and complement, governed by laws such as De Morgan's laws. Venn diagrams provide visual representation of set relationships. Set theory, developed by Georg Cantor, forms the language of modern mathematics and underpins probability, logic and computer science.

Chapter 2

Trigonometric Functions

Trigonometry relates the angles and sides of triangles. The six trigonometric functions — sine, cosine, tangent, cosecant, secant and cotangent — extend to circular functions defined for all real numbers using the unit circle. Identities such as $\sin^2 + \cos^2 = 1$ and the sum and difference formulas enable simplification of expressions and solution of equations. Trigonometric functions model periodic phenomena such as sound waves, tides and oscillations.

Chapter 3

Calculus: Limits and Derivatives

Calculus studies change and accumulation. The limit describes the value a function approaches as the input approaches a point. The derivative, defined as the limit of the difference quotient, measures the instantaneous rate of change and gives the slope of the tangent. Derivatives of common functions follow established rules such as the product, quotient and chain rules. Calculus, developed independently by Newton and Leibniz, is indispensable in physics, engineering, economics and biology.